

1. D
2. C
3. D
4. C
5. C
6. A
7. B
8. C
9. C
10. B
11. D
12. B
13. D
14. C
15. B
16. D
17. True
18. A
19. B
20. C

21. Supervised learning is the process or a way of a machine learning algorithm to learn from labeled input. or labeled data.

27. To find out the outlier in data we can use the following: Visualization method, statistical method and machine learning method.

31. k in k -means algorithm is the number of partitions that you input. Ex: $k=3$ meaning 3 partitions.

25. Predicting clients risk of death in insurance company, studying crowd control.

29. Data normalization is required in k-NN because:

- Data need a normalization to scale data points into the same scale
- After normalization, when data points are on the same scale k-NN will be able to group similar features of data points into target class.

23. Regression analysis is the process of mapping relationship of variables in a dataset.

24. Elucidate two challenges of Machine Learning:

- Accurate Accuracy of predictions
- Not confident on making predictions (Log-loss)

26. Distinguish between Structured and Semi-Structured data:

- Structured Data is like ordinal data. It is sorted and order in sequence.
- Semi Structured Data is like Nominal data. It is in any order.

28. Data pre-processing is the process of cleaning data before it is given to a model.

Pre-processing includes finding null value, finding missing value, removing redundant or unnecessary data from dataset.

32. Hyper parameters of machine learning algorithms:

- k-nn, Linear regression, Logistic regression, k-clustering, Supported Vector Machine.

34. Comment on the performance of classifiers:

First SVM: Mediocre accuracy and precision. Bad recall. Able to separate data points into two classes with an okay confidence. (Log-loss)

Second SVM: Bad accuracy and precision. Terrible recall. Not able to separate data points into two classes. Bad F-1 score and no confidence.

Third SVM: Good precision and accuracy. Able to completely differentiate data points into two classes with its hyper plane. It should have a good F1 score and good log loss.